

Homework 2

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Exercise 1

Consider the $2 \times n$ Chomp game.

- What is a winning strategy for the first player?
- What are the Grundy numbers of the possible positions in the game?

Solution:

- The winning strategy for the first player in the case of the $2 \times n$ chomp game is to first take off the top right most square, then always reestablish a position where only the top right square is missing.

It is clear to see that this reestablishing can always be done in the case of the $2 \times n$ chomp game.

There are two types of moves the second player can play from this position:

- If the second player makes a $2 \times n'$ rectangle then we are back to the starting position.
- If the second player makes a position where the top row is shorter than the bottom row, then the first player takes off just enough of the bottom row's right hand side to make the rectangle with the missing corner again.

Since the first player can maintain this position of the rectangle with the missing corner, we end up in the position where the second player is to play in the L shape position with 3 squares.

By checking all possible moves of the second player we see that the first player wins in all cases.

- Let us number the positions with the tuple (n, m) where n represents how many squares are in the bottom row, and m represents how many squares are in the top row. For example, $(2, 1)$ represents the L shape 3 square position.

By writing out the Grundy number for the first few position we can make an educated guess for a general position's Grundy number:

$$\begin{pmatrix} n \backslash m & 0 & 1 & 2 & 3 & 4 & 5 & 6 \\ 1 & 0 & 1 & & & & & \\ 2 & 1 & 0 & 2 & & & & \\ 3 & 2 & 3 & 0 & 4 & & & \\ 4 & 3 & 2 & 4 & 0 & 5 & & \\ 5 & 4 & 5 & 3 & 6 & 0 & 7 & \\ 6 & 5 & 4 & 6 & 3 & 7 & 0 & 8 \end{pmatrix}$$

We can see that in the diagonals where $d = n - m$ is constant there are two kinds of patterns, the one when d is even and when d is odd.

It remains to be shown what the patterns are and that they truly constitute the Grundy numbering for the $2 \times n$ Chomp game.

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Exercise 2

Let game G be the sum of $\{1, 2\}$ -nim and $\{2, 3\}$ -nim. Compute the Grundy number and the winning first move (if it exists) for starting positions $(2, 2)$, $(4, 4)$, and $(101, 101)$.

Solution: Using the Sprague–Grundy theorem we need only find the Grundy numbers of the single game positions $g_1(2), g_2(2)$, then $g((2, 2)) = g_1(2) \oplus g_2(2)$, and likewise for the other two positions.

It can be showed that $g_1(p) = p(\bmod 3)$ and $g_2(p) = \left\lfloor \frac{p(\bmod 5)}{2} \right\rfloor$. This can be seen by writing out the first few positions and calculating the Grundy numbers, then proving the guess by induction.

The Grundy numbers for the asked positions of the game sum are:

$$\begin{aligned} g((2, 2)) &= 2(\bmod 3) \oplus \left\lfloor \frac{2(\bmod 5)}{2} \right\rfloor = 2 \oplus 1 = 3 \\ g((4, 4)) &= 4(\bmod 3) \oplus \left\lfloor \frac{4(\bmod 5)}{2} \right\rfloor = 1 \oplus 2 = 3 \\ g((101, 101)) &= 101(\bmod 3) \oplus \left\lfloor \frac{101(\bmod 5)}{2} \right\rfloor = 2 \oplus 0 = 2. \end{aligned}$$

Since all of the Grundy numbers for the position pairs are positive, this means that the first player has a winning move in each position. The winning move is the one that takes us to a position where $g((p_1, p_2)) = 0$, that is $g_1(p_1) \oplus g_2(p_2) = 0$.

For position $(2, 2)$ the winning first move is to go to $(1, 2)$, since $1(\bmod 3) \oplus 1 = 1 \oplus 1 = 0$.

For position $(4, 4)$ the winning first move is to go to $(2, 4)$, since $2(\bmod 3) \oplus 2 = 2 \oplus 2 = 0$.

For position $(101, 101)$ the winning first move is to go to $(99, 101)$ since $99(\bmod 3) \oplus 0 = 0 \oplus 0 = 0$.

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